

$$F(k)dk \xrightarrow{\sum} \int F(k)dk = F$$

$$Z = \sum_n e^{-\beta E_n}$$

$$F(k)dk = -k_B T D(k) \ln(Z) dk$$

\* Harmonic Oscill.:  $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2$  /  $[a, a^\dagger] = i\hbar$   
 $E_n = (\frac{1}{2} + n)\hbar\omega$  (1D)

$$Z = \sum_n e^{-\beta E_n} \rightarrow \frac{1}{2}\hbar\omega \text{ (zero pt en.)}$$

$$E_0 = \frac{1}{2}\hbar\omega \leadsto \text{"zero"}$$

$$E_1 = \frac{3}{2}\hbar\omega \leadsto \hbar\omega = \tilde{E}_1$$

\* Stat physics:  $\tilde{E}_n = n\hbar\omega$

$$Z = \sum_n e^{-\beta \tilde{E}_n} = \sum_n e^{-\beta n\hbar\omega} \quad * \text{Single}$$

$$Z = \sum_n e^{-\beta E_n}$$

$\leftarrow$  over all particles in 3D

$$\sum_n = \sum_{n_x} \sum_{n_y} \sum_{n_z} \rightarrow \int d^3n$$

$D(k)$ : Density

$$\rightarrow \int D(k) dk$$

\* Particle in a 3D Box: length  $L$

$$V = L^3, \quad \hat{H} = \frac{\hat{p}^2}{2m}$$

$$\psi(0) = 0, \quad \psi(L) = 0$$

$$\hat{H}\psi = E\psi :$$



$$\psi(x, y, z) = \sin\left(\frac{n_x}{2L}x\right) \sin\left(\frac{n_y}{2L}y\right) \sin\left(\frac{n_z}{2L}z\right)$$

$$\lambda_n = \frac{n}{2L} \Rightarrow p_x = \frac{\hbar n_x}{2L}, \quad p_y = \frac{\hbar n_y}{2L}, \quad p_z$$

$$E = \frac{|\vec{p}|^2}{2m} = \frac{\hbar^2}{(2L)^2 2m} (n_x^2 + n_y^2 + n_z^2)$$

$$\vec{p} = (p_x, p_y, p_z)$$

Sphere:  $x^2 + y^2 + z^2 = R^2$

$$\underbrace{\left( \frac{(2L)^2 2m}{\hbar^2} \right)}_{R^2} E = n_x^2 + n_y^2 + n_z^2$$

- Energy modes are constrained to a sphere:

$$p_x = \hbar n_x / 2L$$

$$\boxed{k_x = 2\pi n_x / L}$$

$$\int d^3n = \int dn_x dn_y dn_z \hookrightarrow$$

$$\rightarrow dk = \frac{2\pi}{L} dn \Rightarrow dn = \frac{L}{2\pi} dk$$

$$\dots \frac{L^3}{2\pi} \int d^3k$$

$$\int d^3n = (2\pi)^3 \int d^3k$$

Box:

$$V = L^3$$

$$\sum_n \longrightarrow \int d^3n = \frac{V}{(2\pi)^3} \int d^3k$$

$\int d^3k$  Spherical coord.:

$$\int |k|^2 \sin(\theta) dk d\theta d\phi$$

$$(r, \theta, \phi) : \int r^2 \sin(\theta) dr d\theta d\phi$$

$$= \int_0^\infty |k|^2 dk (4\pi)$$

$$= \left( \int_0^{2\pi} d\phi \right) \left( \int_0^\pi \sin(\theta) d\theta \right) \left( \int_0^\infty |k|^2 dk \right)$$

$$\frac{V}{(2\pi)^3} 4\pi \cdot \int_0^\infty |k|^2 dk$$

$$2 \int_0^\infty \frac{V}{2\pi^2} |k|^2 dk = \sum_n$$

Polarization  
of light

Recall:  $\int D(k) dk$

$$* \boxed{D(k) = \frac{V}{\pi^2} |k|^2}$$

$$\int F(k) dk = F = -k_B T \int_0^\infty \frac{V}{\pi^2} |k|^2 \underline{f_1(z)} dk$$

$$Z = \sum_n e^{-\beta E_n} = \sum e^{-\beta \hbar \omega}$$

$$W = ck$$

$$Z = \sum_n e^{-\beta ck} = \frac{1}{1 - e^{-\beta ck}}$$

$$* \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad | \quad 0 < |x| < 1$$

(proof in notes)

$$\ln(Z(k)) = \ln\left(\frac{1}{1 - e^{-\beta ck}}\right)$$

$$= -\ln(1 - e^{-\beta ck})$$

$$F = \left(-\frac{k_B T V}{\pi^2}\right) \int |k|^2 \left(-\ln(1 - e^{-\beta ck})\right) dk$$

$$F = \frac{k_B T V}{\pi^2} \int k^2 \ln(1 - e^{-\beta ck}) dk$$

$$F = -k_B T \ln(Z)$$

\* In QM: each state  $|n\rangle$  has a prob.

$$P(n) = e^{-\beta E_n} / Z \quad (\text{prob of being in } |n\rangle)$$

$$1 = \sum_n P(n) = \sum_n \frac{e^{-\beta E_n}}{Z} = \frac{1}{Z} \sum_n e^{-\beta E_n}$$

$$= \frac{Z}{Z} = 1$$

\* Energy:  $U_{int} = \overline{E} = \langle E \rangle$   
(notation)

in QM:  $\langle \psi | \hat{H} | \psi \rangle = \langle E \rangle$

$$\langle E \rangle = \sum_n E_n P(n)$$

$$= \sum_n E_n \frac{e^{-\beta E_n}}{Z} = \frac{1}{Z} \sum_n E_n e^{-\beta E_n}$$

$$E_n e^{-\beta E_n} = - \frac{\partial}{\partial \beta} e^{-\beta E_n}$$

$$\langle E \rangle = \frac{1}{Z} \sum_n \left( - \frac{\partial}{\partial \beta} e^{-\beta E_n} \right)$$

$$\langle E \rangle = - \frac{1}{Z} \frac{\partial}{\partial \beta} \left( \underbrace{\sum_n e^{-\beta E_n}}_Z \right)$$

$$\langle E_n \rangle = U_{int} = - \frac{1}{Z} \frac{\partial Z}{\partial \beta}$$

lii)  $S = k_B (\ln(Z) + \beta \langle E \rangle)$

$$F = -k_B T \ln(Z)$$

$$\beta = \frac{1}{k_B T}$$

$$= \beta F = \ln(Z)$$

$$\Rightarrow S = k_B (-\beta F + \beta \langle E \rangle)$$

$$= k_B \beta (-F + \langle E \rangle)$$

$$= \frac{k_B}{k_B T} (-F + \langle E \rangle)$$

$$S = \frac{1}{T} (-F + \langle E \rangle)$$

$$\text{also: } S = -\frac{\partial F}{\partial T}$$

$$\langle E \rangle = U_{int} = -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{\partial}{\partial \beta} (\ln(Z))$$

$$\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \frac{\partial \ln(Z)}{\partial \beta}$$

$$F = -\beta^{-1} \ln(Z)$$

$$-\beta F = \ln(Z)$$

$$-\frac{\partial}{\partial \beta} (\beta F) = \frac{\partial}{\partial \beta} (\ln(Z))$$



$$\beta = \frac{1}{k_B T}$$

$$F = \frac{V}{\beta \pi^2} \int_0^\infty k^2 \ln(1 - e^{-\beta \hbar c k}) dk$$

$$-\frac{\partial}{\partial \beta} (\beta F) = -\frac{\partial}{\partial \beta} \left( \frac{V}{\pi^2} \int_0^\infty k^2 \ln(1 - e^{-\beta \hbar c k}) dk \right)$$

$$RHS = -\frac{V}{\pi^2} \int_0^\infty k^2 \frac{1}{1 - e^{-\beta \hbar c k}} \left( \frac{\hbar c k}{\dots} \right) e^{-\beta \hbar c k}$$

$$\langle E \rangle = -\frac{V \hbar c}{\pi^2} \int_0^\infty k^3 \frac{e^{-\beta \hbar c k}}{1 - e^{-\beta \hbar c k}}$$

Solve:  $S = \frac{1}{T} (-F + \langle E \rangle)$

$\uparrow$                        $\uparrow$   
 from (a)

